

Sai Rithwik Kukunuri

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EDUCATION

Arizona State University

B.S. Computer Science (Barrett Honors College)
Minor: Data Science | Cumulative GPA: 4.00/4.00

Aug 2024 – May 2027

Tempe, Arizona

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL

Frontend: React, Next.js, JavaFX, HTML, CSS, Figma

Backend & APIs: FastAPI, Flask, Spring Boot, REST APIs, PostgreSQL, MySQL, MongoDB, H2

AI & Data: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, SAM 2, Qdrant, RAG Pipelines, Groq

Cloud & Tools: Git, GitHub, GitHub Actions, Docker, CI/CD, AWS, Azure, Linux, Jira, JUnit, UML, Agile/Scrum

EXPERIENCE

Undergraduate Teaching Assistant

May 2026 – Jul 2026

Arizona State University

Tempe, Arizona

- Supported students in CSE 360 with Java, JavaFX, SQL, GitHub, Jira, JUnit, UML, debugging, testing, and software engineering workflows.
- Diagnosed implementation, database, testing, and version-control issues through office hours, code reviews, and structured technical troubleshooting.
- Created technical documentation and collaborated with instructors and course staff to communicate requirements, track student concerns, and resolve recurring technical issues.

AI-Powered Question Answering for Medical Procedures

Jan 2026 – Present

Undergraduate Researcher, TEAL Lab, ASU

Tempe, Arizona

- Built automated workflows for processing text, images, voice, and video frames using validation, embeddings, semantic search, and timestamped responses.
- Developed a full-stack medical Q&A application using Python, React, Flask, Qdrant, and REST APIs to retrieve relevant video segments.
- Translated research requirements into iterative improvements and documented APIs, testing, and results, reducing latency by 30%.

IPv4 IPID Traffic Modeling and Network Security

Jan 2026 – Present

Undergraduate Research Assistant, Biodesign Institute, ASU

Tempe, Arizona

- Developed Python workflows for IPv4 traffic modeling, data processing, validation, testing, and automated reporting.
- Collected and analyzed network traffic using NumPy and SciPy to evaluate IPID collision risk, attack predictability, and bursty traffic patterns.
- Maintained reproducible scripts and technical documentation while reporting findings showing up to 63× higher vulnerability under bursty traffic.

PROJECTS

Stepwise - Interactive Algorithm Visualizer | Astro, TypeScript, React, D3.js, Python, C++

[GitHub](#)

- Developed an interactive platform for visualizing sorting, searching, recursion, trees, graphs, and dynamic programming algorithms.
- Built synchronized visualizations and code execution views that highlight each algorithmic step across Python and C++ implementations.
- Designed reusable visualization components using React and D3.js to illustrate state transitions, data structures, and execution flow.

GitHub AI Code Review Agent | Python, FastAPI, LangChain, GitHub API, Docker, GitHub Actions

- Built an AI software engineering assistant that reads GitHub issues, identifies relevant code files, and suggests step-by-step fixes.
- Added pull request review features to detect bugs, missing tests, unclear logic, and potential security risks.
- Integrated linting, automated tests, and static analysis checks to flag changes that increase complexity or reduce code quality.

LEADERSHIP

Google Developer Student Club

Sept 2025 – Present

Graphic Design Officer

Tempe, Arizona

- Designed and delivered visual content for developer workshops improving communication and engagement for a 2,000+ member community.
- Recognized as a 2026 Pitchfork Awards Finalist for contributions to student organization initiatives.

Software Developers Association

Jan 2025 – Present

Marketing Officer

Tempe, Arizona

- Promote workshops and developer events for a 3,000+ member community and design visual content to increase participation and outreach.
- Develop and maintain the organization's website features using React, TypeScript, and Flask improving usability and system functionality.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

Amazon Web Services, 2026

AWS Certified AI Practitioner

Amazon Web Services, 2026

Microsoft Certified: Power Platform Fundamentals

Microsoft, 2026